# Gregory Polo

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## EDUCATION

**Syracuse University**, College of Engineering and Computer Science **GPA: 3.54/4.00**
*Bachelor of Science in Computer Engineering* *Expected May 2028*
Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

## ENGINEERING APPLICATIONS

**Educational Cybersecurity Game** | *Ren'Py (Python)* *Aug 2026 – Present*

- Chose to build a narrative training game for a cybersecurity course project to help people in cybersecurity roles navigate and respond to typical real-world attacks, working with a 6-person team to plan and build it.
- Storyboarding 3 scenarios covering phishing, cross-site scripting (XSS), HTML injection, and password attacks, and composing most of the original soundtrack that plays throughout the game, alongside the team.
- Building toward a playable educational tool that teaches players in multiple roles to recognize, respond to, and defend against common cyberattacks, developed in Python with Ren'Py as a team course project.

**Arduino Sensor Interfacing** | *Arduino*, *C++*, *Python*, *Digilent WaveForms* *Mar 2026*

- Tasked with measuring a joystick and a swinging pendulum with an Arduino and Analog Discovery, then building theoretical models to match the recorded pendulum data and reporting the results with a lab partner.
- Recorded the pendulum's voltage over time and mapped each reading to an angle with $(V - 1.65)/3.3 \times 360°$ in Python, then modeled the swing from a 142.7° release with a linear state-space model and an RK45 nonlinear ODE.
- Confirmed the 1.65 V reference matched across theory, the Arduino serial monitor, and Analog Discovery, and the nonlinear model matched the 17-second decay almost exactly, closer than the linear model did.

**4-Bit CPU on Intel MAX 10 FPGA** | *VHDL*, *Quartus Prime*, *Analog Discovery* *Feb 2026*

- Tasked with building a 4-bit CPU from scratch in VHDL on the Intel MAX 10 DE10-Lite FPGA that could run LDA and ADDA instructions from a 16-address ROM, working with a lab partner and a provided component set.
- Wired seven components (instruction and operand registers, PC, ALU, accumulator, ROM, FSM) plus six hex displays, and coded a 6-state FSM that completes each instruction in 4 clock cycles and drives every control signal.
- Delivered a working CPU using only 25 logic elements and 22 registers that ran the 8-instruction ROM program at 1 Hz and a second program I wrote (A = 1, 3, 7, 9, A, 8, 6, 2), both demonstrated to the instructor.

## LEADERSHIP | VOLUNTEER | EXTRACURRICULAR

**The William J. and Dorothy K. O'Neill Foundation** **Cleveland, OH**
*Youth Board Member* *2016 – Present*

- Serve on the youth board of a Cleveland-based foundation, tasked with identifying community needs and helping guide where grant funding is directed, a role held since 2016 and now in its tenth year.
- Lead community summits with multiple nonprofit organizations to assess local needs, and connect them with peer nonprofits and foundations to share problem-solving strategies and learn from each other.
- Helped shape the foundation's funding priorities and grantmaking decisions, directing funding to nonprofits that address a broad range of community needs across the greater Cleveland area over the course of ten years.

**Windermere Preparatory School Summer Camp** **Windermere, FL**
*Camp Counselor* *Jun – Aug 2022, 2023*

- Responsible for the safety and daily supervision of 30 campers ages 5–10 across two consecutive summers at a private school's summer camp program in Windermere, Florida, from June through August each summer.
- Planned and led outdoor and arts-and-crafts activities, adapting instructions to fit a wide range of ages, attention spans, and skill levels so every camper could take part and stay engaged throughout the day.
- Maintained a safe, positive environment for every camper throughout daily programming across both summers, earning a return position for a second summer after completing the first season at camp.

## TECHNICAL SKILLS

**Hardware & Tools:** Oscilloscope, Multimeter, Digilent WaveForms, Arduino, Intel MAX 10 FPGA, Intel Quartus Prime
**Programming Languages:** C, C++, VHDL, Python (NumPy, SciPy, pandas), Java, Haskell, Bash
**Concepts:** Digital Logic Design, Computer Architecture, Object-Oriented Programming, Data Structures, Linux
**Spoken Languages:** English, Spanish (conversational)